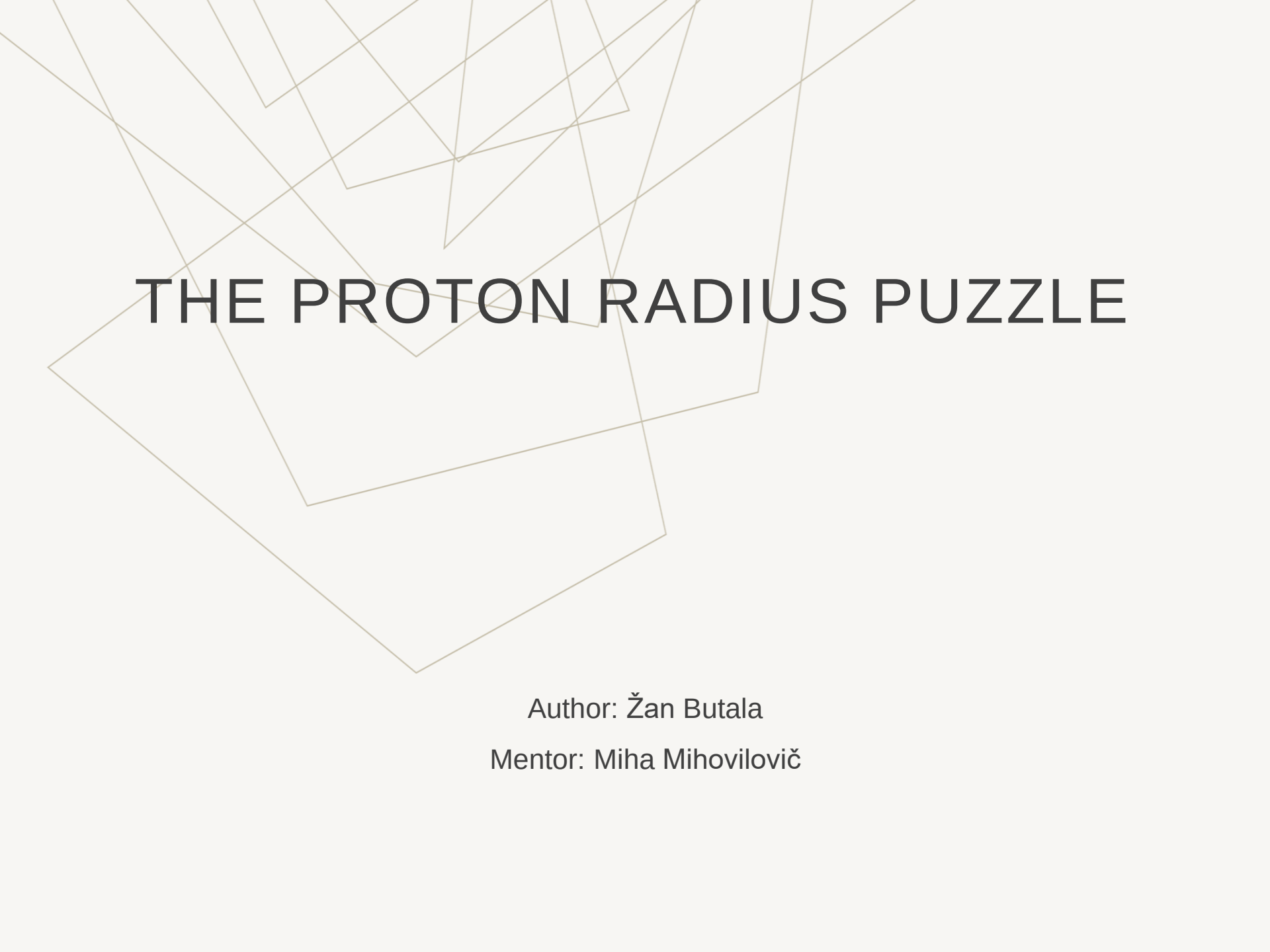




THE PROTON RADIUS PUZZLE

Author: Žan Butala

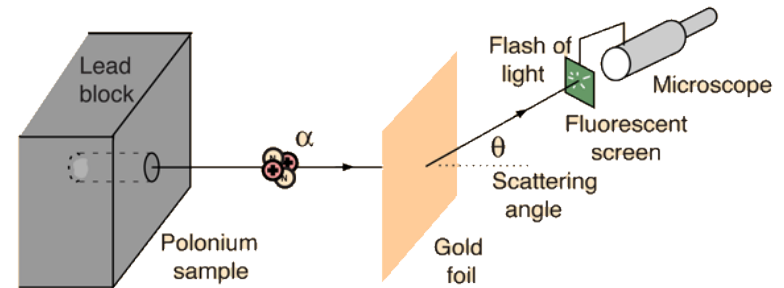
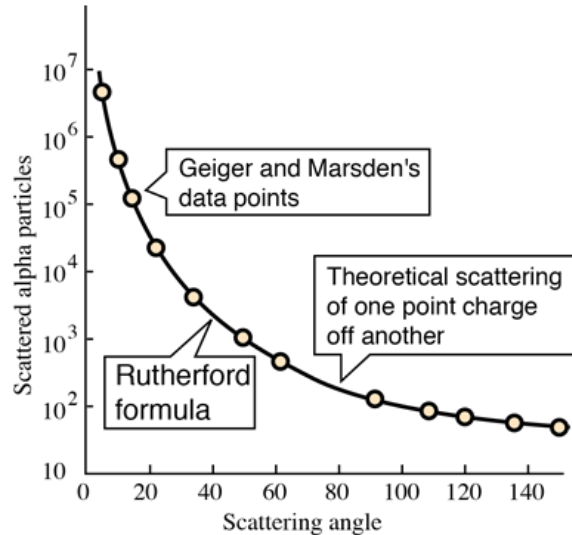
Mentor: Miha Mihovilovič

OUTLINE

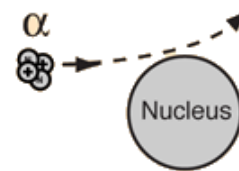
- Introduction
- Elastic scattering
- Hydrogen spectroscopy
- Results and the “puzzle”
- Possible solutions

DISCOVERY OF PROTON'S FINITE SIZE

- Rutherford scattering experiment (1911):
Charge concentrated in small, compact area

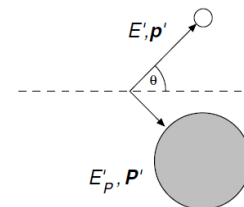
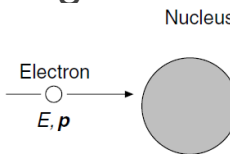


$$\frac{d\sigma}{d\Omega} = \frac{Z^2 \alpha^2}{4E^2 \sin^4 \frac{\theta}{2}}$$



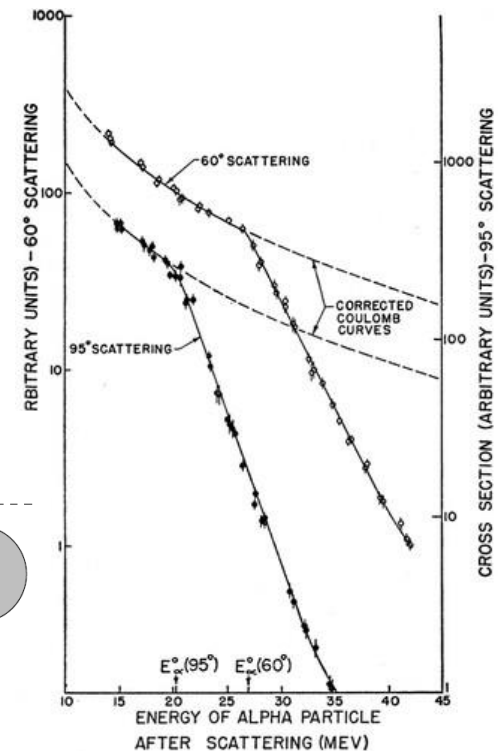
- Increasing energy and scattering off other atoms (1919):

Hydrogen nuclei detected as b...



- Otto Stern (1933):

Anomalous magnetic moment of proton 2.5-times larger than predicted by Dirac eq.



DIFFERENTIAL CROSS-SECTION

- Mott cross-section

$$\left(\frac{d\sigma}{d\Omega}\right)_{Mott} = \frac{\alpha^2}{4E^2 \sin^4 \frac{\theta}{2}} \frac{E'}{E} \left(1 - \beta^2 \sin^2 \frac{\theta}{2}\right)$$

- Hoffstadter (1956):

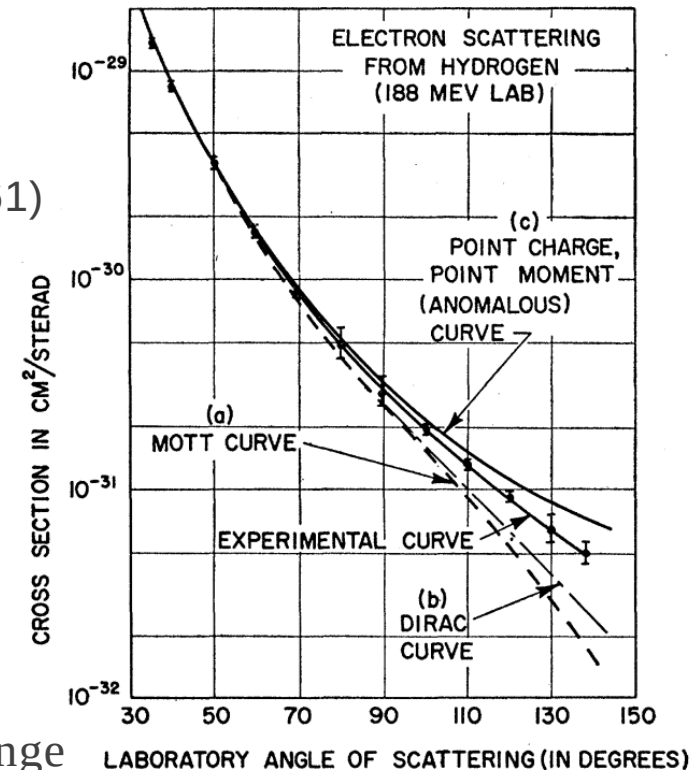
Point-like proton > Exp. > Mott, (Nobel Prize 1961)

$$\left(\frac{d\sigma}{d\Omega}\right)_{exp.} = \left(\frac{d\sigma}{d\Omega}\right)_{Mott} |F(q^2)|^2$$

- Charge density and form factor:

$$\rho(x) \longrightarrow F(\mathbf{q}) = \int \rho(x) e^{i\mathbf{q} \cdot \mathbf{x}} d\mathbf{x}$$

- Exp. $\longrightarrow F(\mathbf{q}) \longrightarrow \rho(x)$
- In practice F can be measured over limited range of q (finite beam energy, rapid drop of cross-section)



DIFFERENTIAL CROSS-SECTION

- Charge density and form factor: $\rho(x) \longrightarrow F(\mathbf{q}) = \int \rho(x) e^{i\mathbf{q} \cdot \mathbf{x}} d\mathbf{x}$

$$\begin{aligned} F(\mathbf{q}) &= \int (1 + i\mathbf{q} \cdot \mathbf{x} - \frac{1}{2}(\mathbf{q} \cdot \mathbf{x})^2 - \frac{i}{3!}(\mathbf{q} \cdot \mathbf{x})^3 + \dots) \rho(\mathbf{x}) d\mathbf{x} \\ &= \int (1 + iqr \cos \theta - \frac{1}{2}q^2 r^2 \cos^2 \theta - \dots) \rho(r) r^2 d\cos \theta d\phi dr \\ &= 1 - \frac{1}{6}|\mathbf{q}|^2 \langle r^2 \rangle + \dots \end{aligned}$$

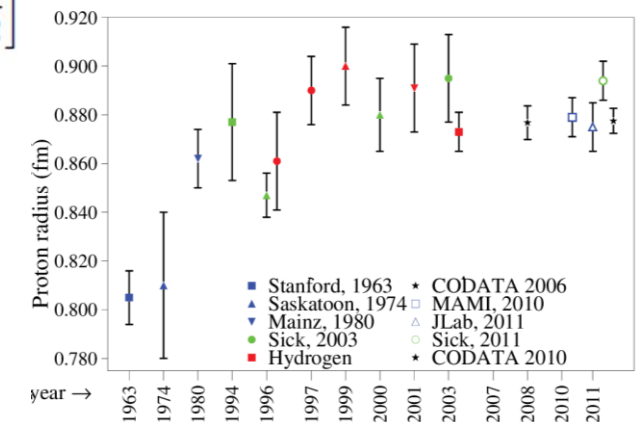
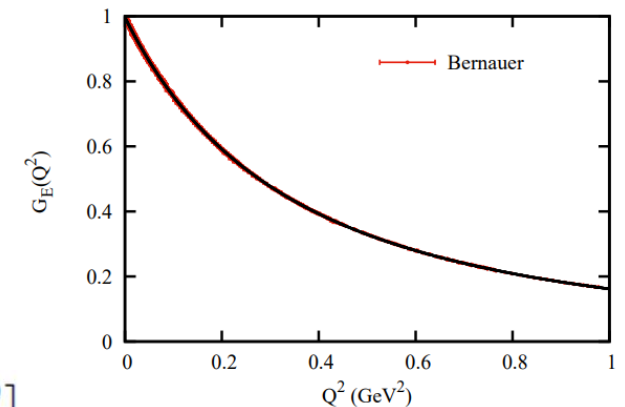
- Charge radius: $R_p^2 \equiv \langle r^2 \rangle = -6 \left. \frac{dF(q^2)}{dq^2} \right|_{q^2=0}$

- Proton's spin, interaction due to electric and magnetic component $\rightarrow$ Two form factors:

$$\left(\frac{d\sigma}{d\Omega} \right)_{lab} = \left(\frac{\alpha^2}{4E^2 \sin^4 \frac{\theta}{2}} \right) \frac{E'}{E} \left[\frac{G_E^2 + \tau G_M^2}{1 + \tau} \cos^2 \frac{\theta}{2} - 2\tau G_M^2 \sin^2 \frac{\theta}{2} \right]$$

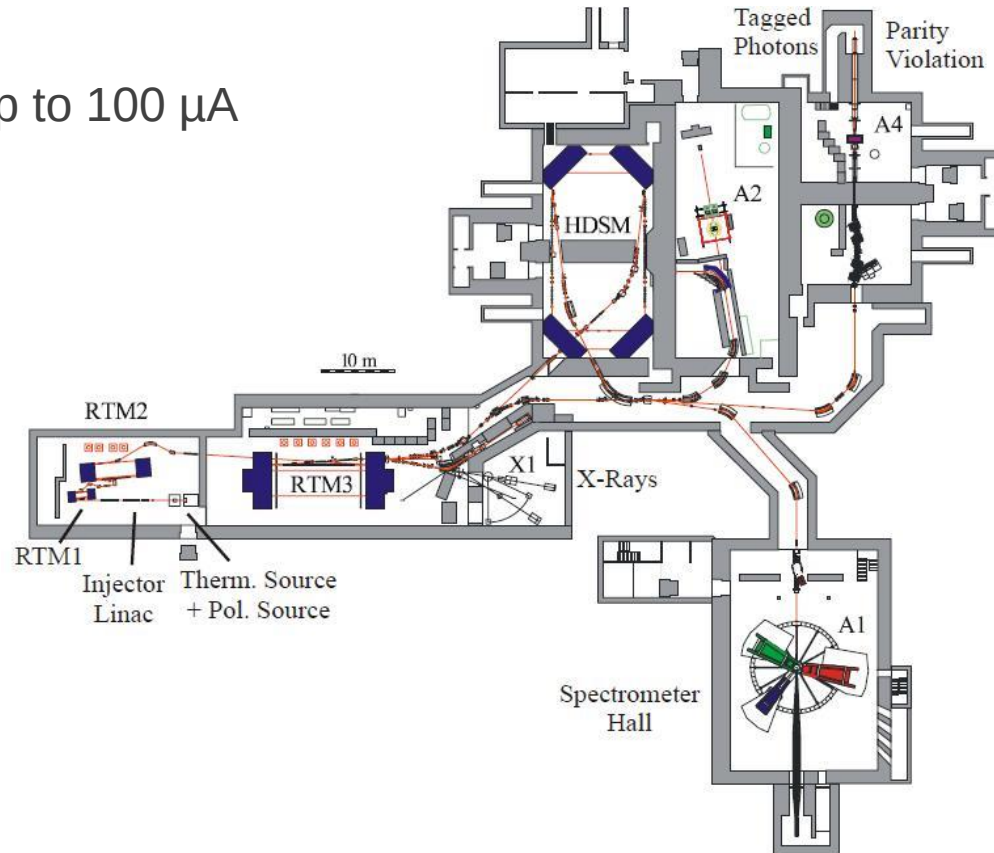
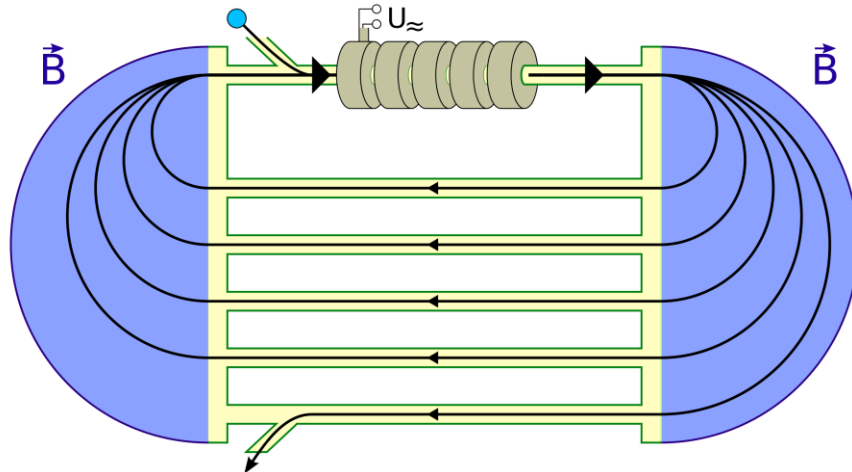
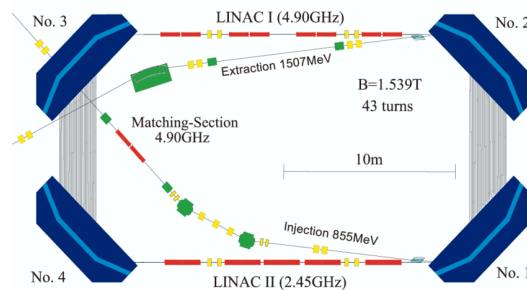
$$R_E^2 = -6 \left. \frac{dG_E}{dQ^2} \right|_{Q^2=0}, \quad R_M^2 = -6 \left. \frac{dG_M}{dQ^2} \right|_{Q^2=0}.$$

- Hand et. al. (1963):
First extraction of proton charge radius



SCATTERING EXPERIMENT

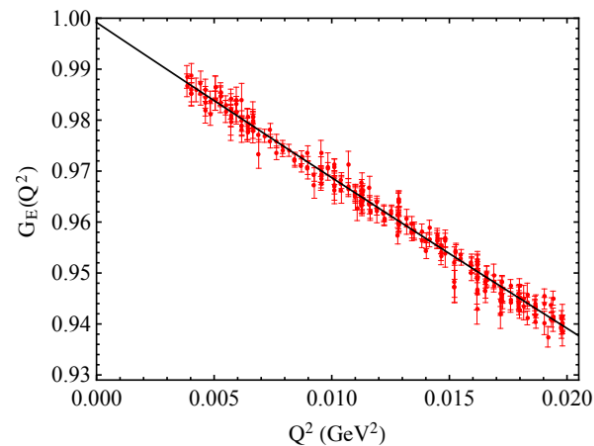
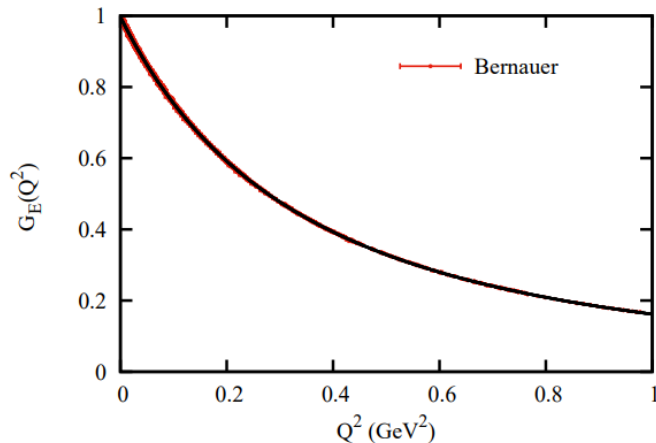
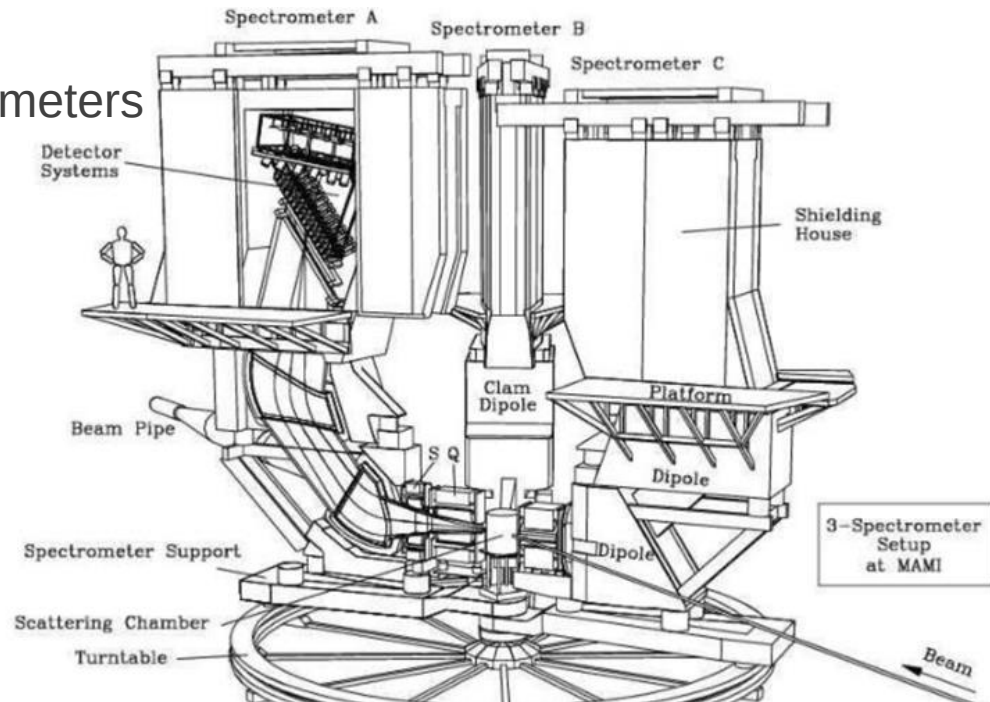
- Bernauer et. al., 2010, Johannes Gutenberg University, Mainz
- Accelerator:
 - Electron beam
 - Energies up to 1.6 GeV
 - Continuous wave, current up to 100 μA
 - Polarised or unpolarised



SCATTERING EXPERIMENT

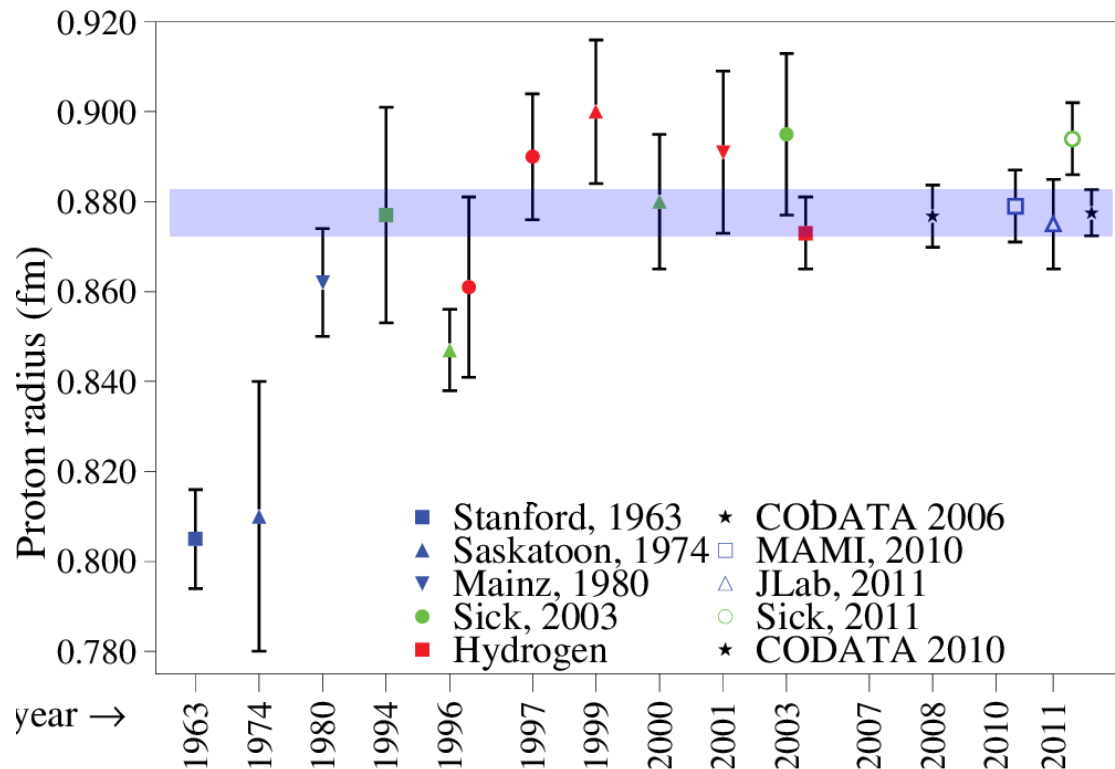
- Experimental Hall A1:
 - 3 high-res. magnetic spectrometers
 - Detector system
 - Target cell, cooled hydrogen

- Results:
 - Electric form factor
 - Charge radius
$$R_p = 0.8795 \pm 0.001) \text{ fm}$$



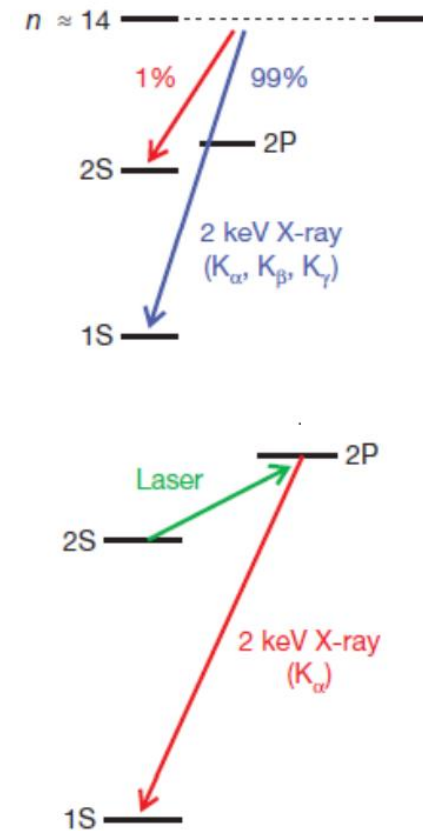
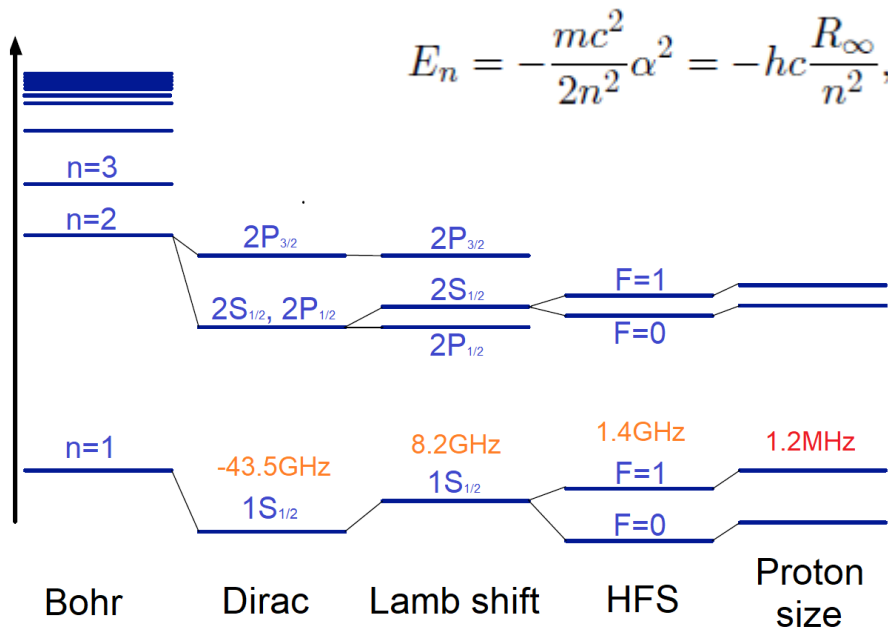
OTHER RESULTS

- Converged to 0.88 fm
- Committee on Data of the International Science Council (CODATA): Recommended values of the Fundamental Physical Constants
 - Average [fm]: 0.879 ± 0.007



HYDROGEN SPECTROSCOPY

- Measuring energy differences between bound states
- Comparing to the theoretical calculations to extract proton radius



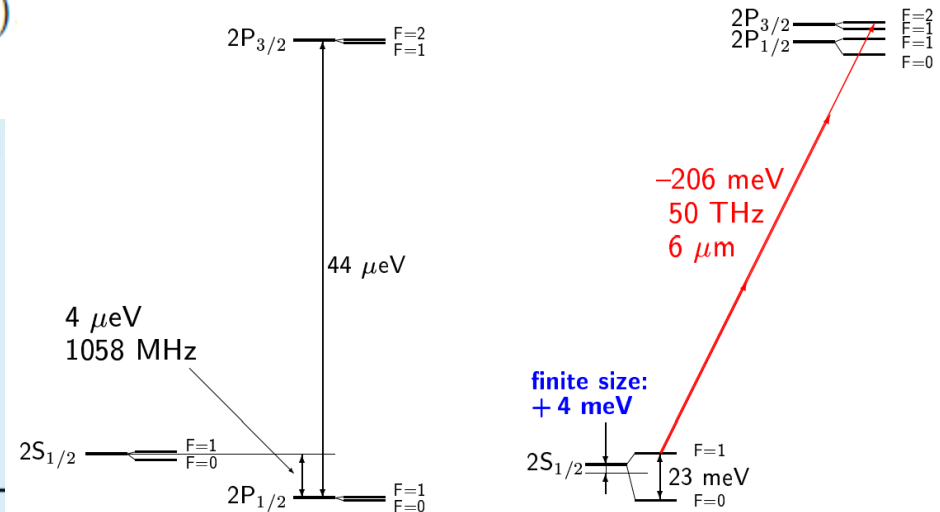
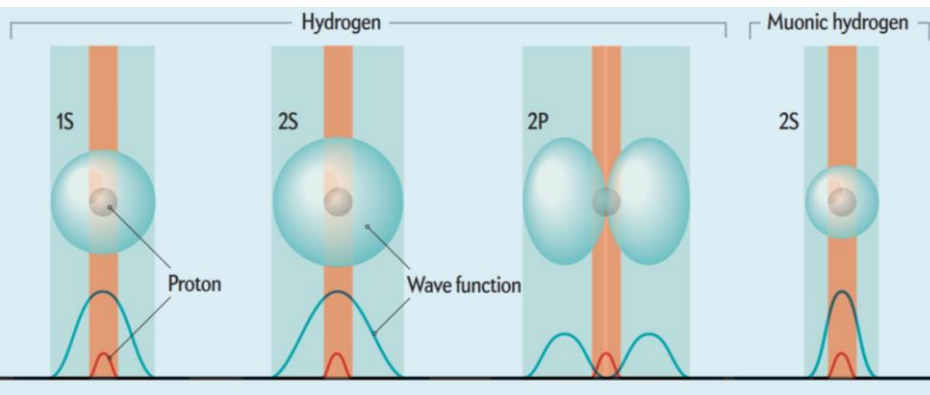
HYDROGEN SPECTROSCOPY

- Dirac equation, splitting of $n > 1$ states:

$$E_{nj} = mc^2 + m_r c^2 (f(n, j) - 1), \quad m_r = \frac{mM}{m + M}$$

- Lamb shift: $\Delta E_{nlj}^{Lamb} = \frac{4mc^2}{3\pi n^3} \alpha^5 \left[L_{nl} + [19/30 - 2 \ln \alpha] \delta_{l0} \pm \frac{3}{4} \frac{1}{(2j+1)(2l+1)} (1 - \delta_{l0}) \right]$
- Finite size correction:

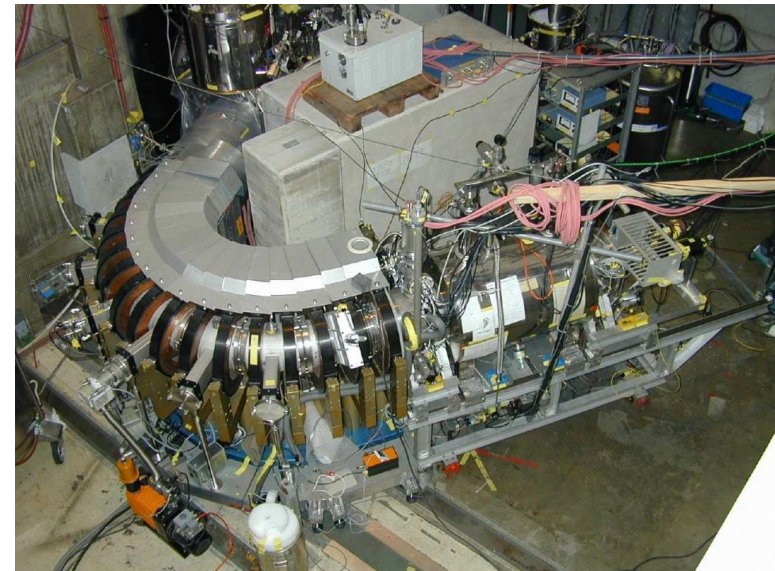
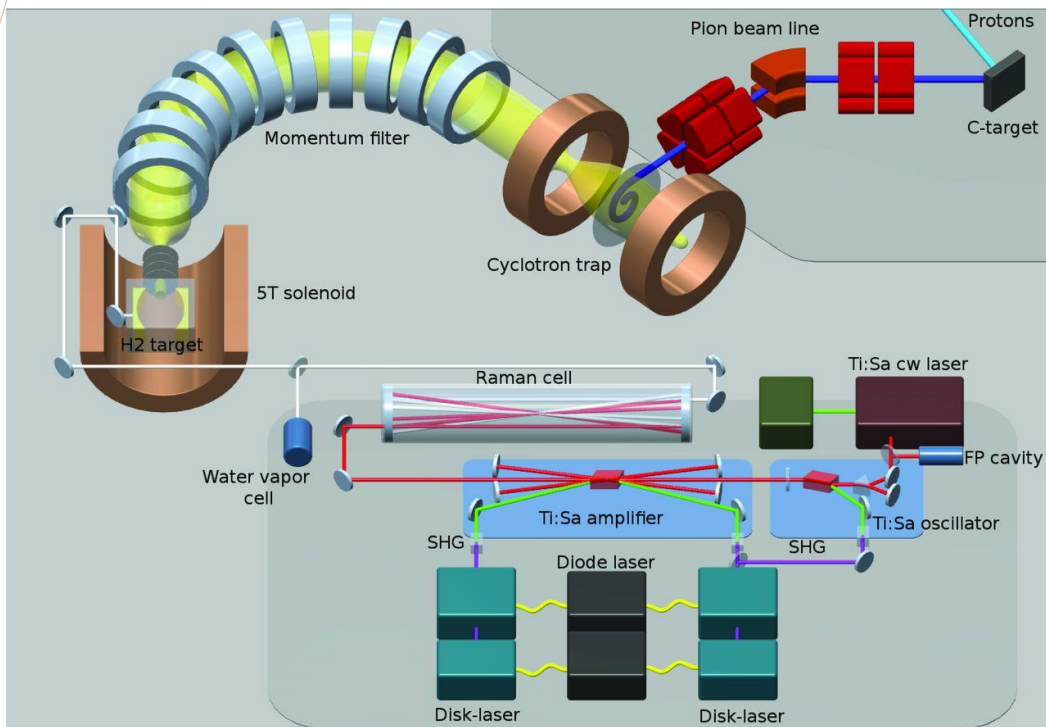
$$\Delta E_V = -\frac{2}{3} \frac{\alpha^4}{n^3} m_r^3 c^6 \left(\frac{R_p}{\hbar c} \right)^2 + \mathcal{O}(R_p^3)$$



MEASUREMENT OF LAMB SHIFT

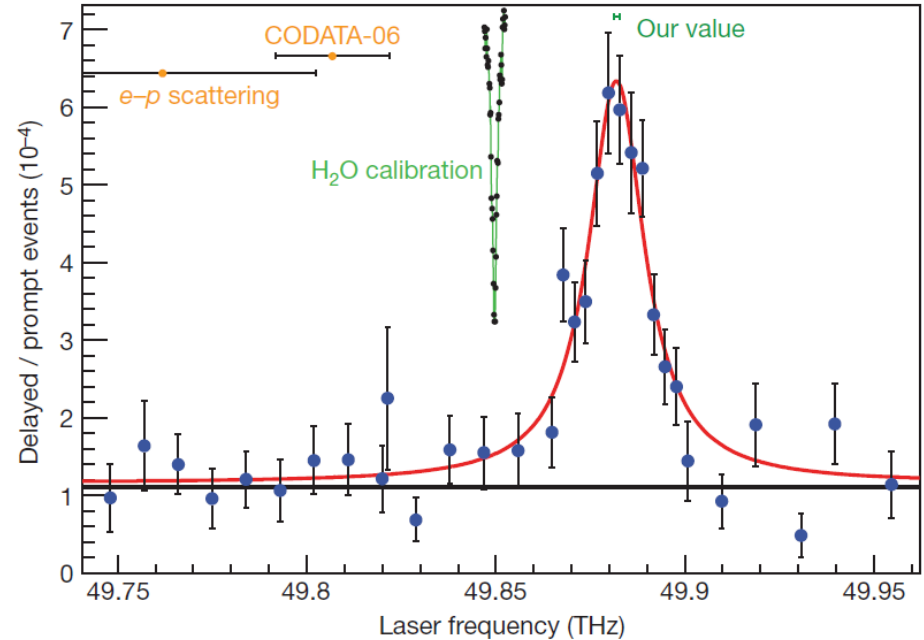
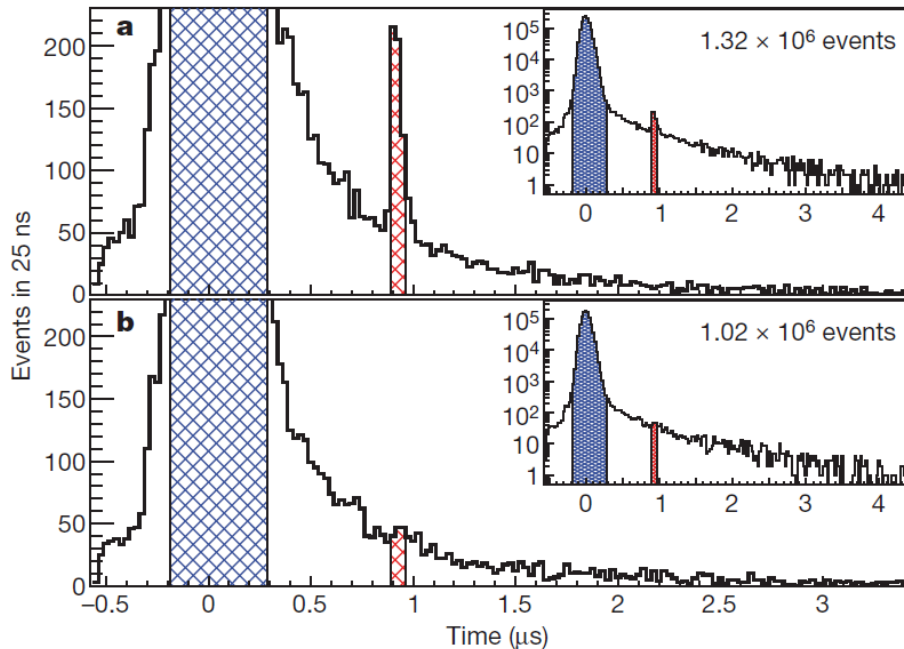
Muonic hydrogen Lamb shift experiment (Paul Scherrer Institute):

- Proton beam to carbon-based target → pions
- Pions decay mostly to muons and neutrinos
- Muons captured in hydrogen atoms
- Transition 2S-2P with laser
- Decay 2P-1S by emitting photon



MEASUREMENT OF LAMB SHIFT

- Detection of emitted photons:

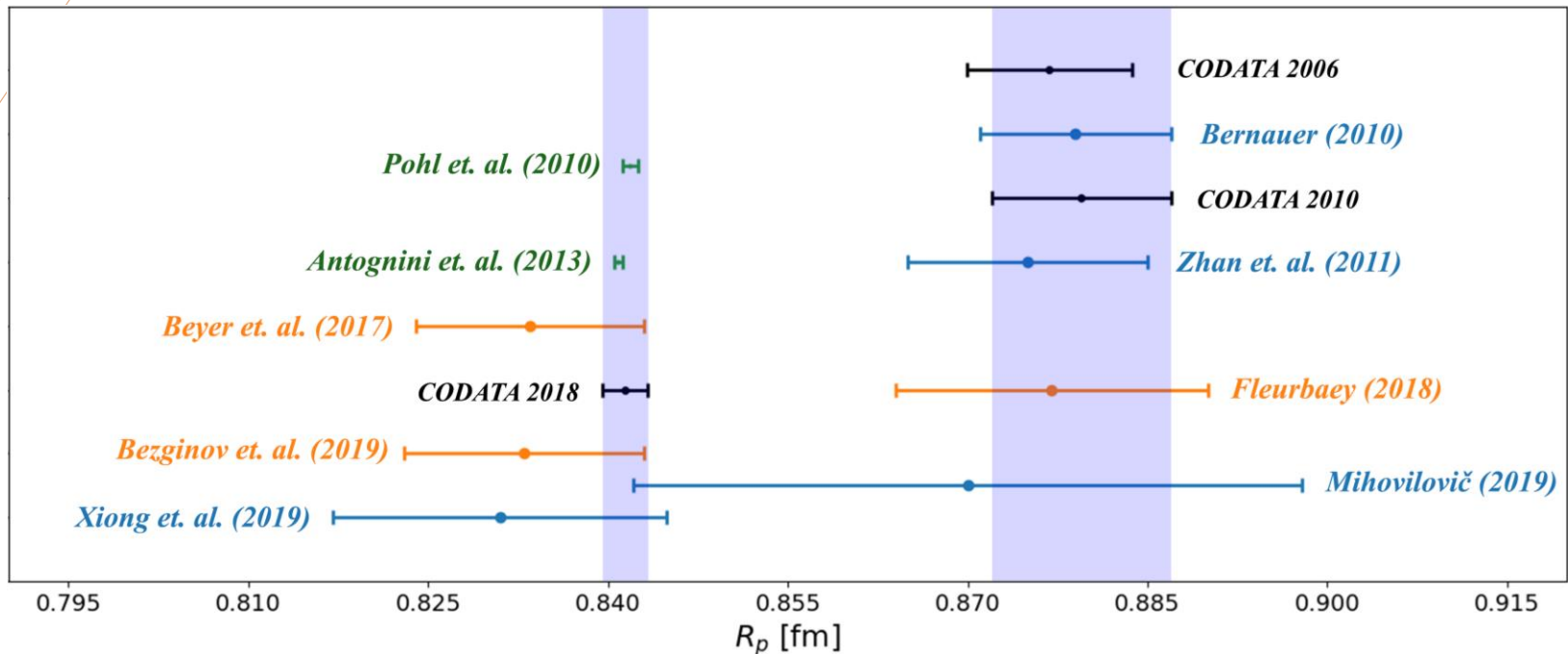


- Parametrization with proton radius:

$$\Delta E [\text{meV}] = 209.9779(49) - 5.2262R_p^2 + 0.0347R_p^3,$$

- Final result [fm]: 0.8418 ± 0.0007

THE PROTON RADIUS PUZZLE

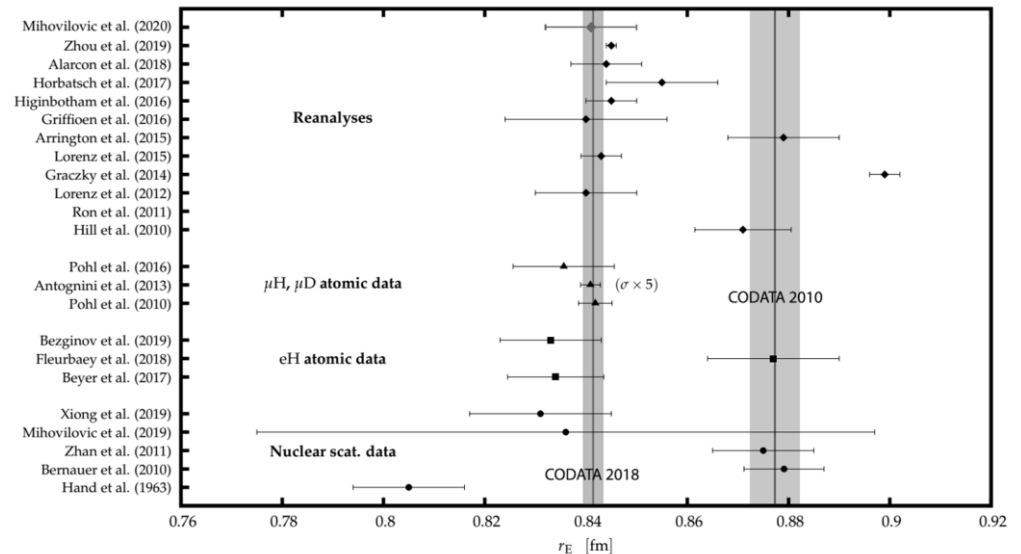
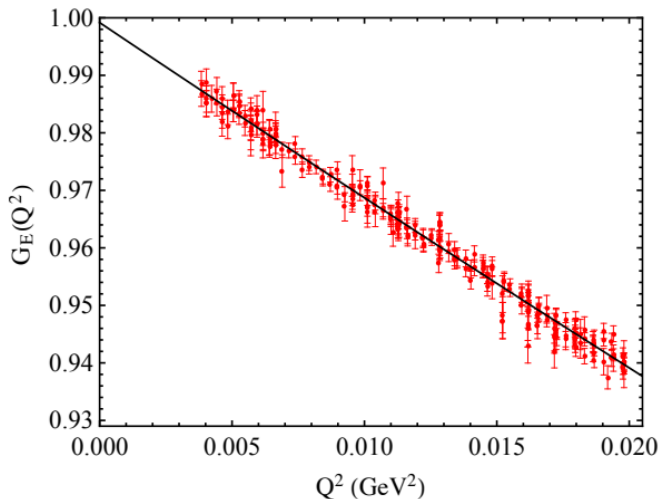


- CODATA 2018:
 - Proton charge radius: $0.8791(79) \text{ fm} \longrightarrow 0.8414(19) \text{ fm}$
 - Rydberg constant: $10\,973\,731.568\,016(49) \text{ m}^{-1} \longrightarrow 10\,973\,731.568\,539(55) \text{ m}^{-1}$

$$E_n \approx R_\infty \left(-\frac{1}{n^2} + \frac{E_{LS}}{n^3} + \dots \right) \quad R_\infty = \frac{m_e e^4}{8\varepsilon_0^2 h^3 c} = 10973731.568160(21) \text{ m}^{-1}$$

POSSIBLE SOLUTIONS

- Spectroscopy and Scattering experiments measuring different quantities:
G. A. Miller: Defining the proton radius: A unified treatment
- Incomplete QED corrections in spectrometry:
U. D. Jentschura: Lamb Shift in Muonic Hydrogen. I. Verification and Update of Theoretical Predictions
- Extrapolation problem and parametric dependency of the form factor models:
K. Griffioen, C. Carlson, S. Maddox: Are Electron Scattering Data Consistent with a Small Proton Radius?



POSSIBLE SOLUTIONS

- Additional corrections to the muonic Lamb shift due to proton structure:
C. E. Carlson, V. Nazaryan, K. Griffioen: Proton-structure corrections to hyperfine splitting in muonic hydrogen
- Higher moments of the charge distribution are needed in the extraction of charge radius:
M. O. Distler, J. C. Bernauer, T. Walcher: The RMS charge radius of the proton and Zemach moments
- BSM Physics, violation of Lepton Universality:
V. Barger et. al.: Proton size anomaly
C. E. Carlson: Extending theories on muon-specific interactions
- Further experiments:
Muonic Scattering Experiment (MUSE) – IN PROGRESS
PRad II

SUMMARY

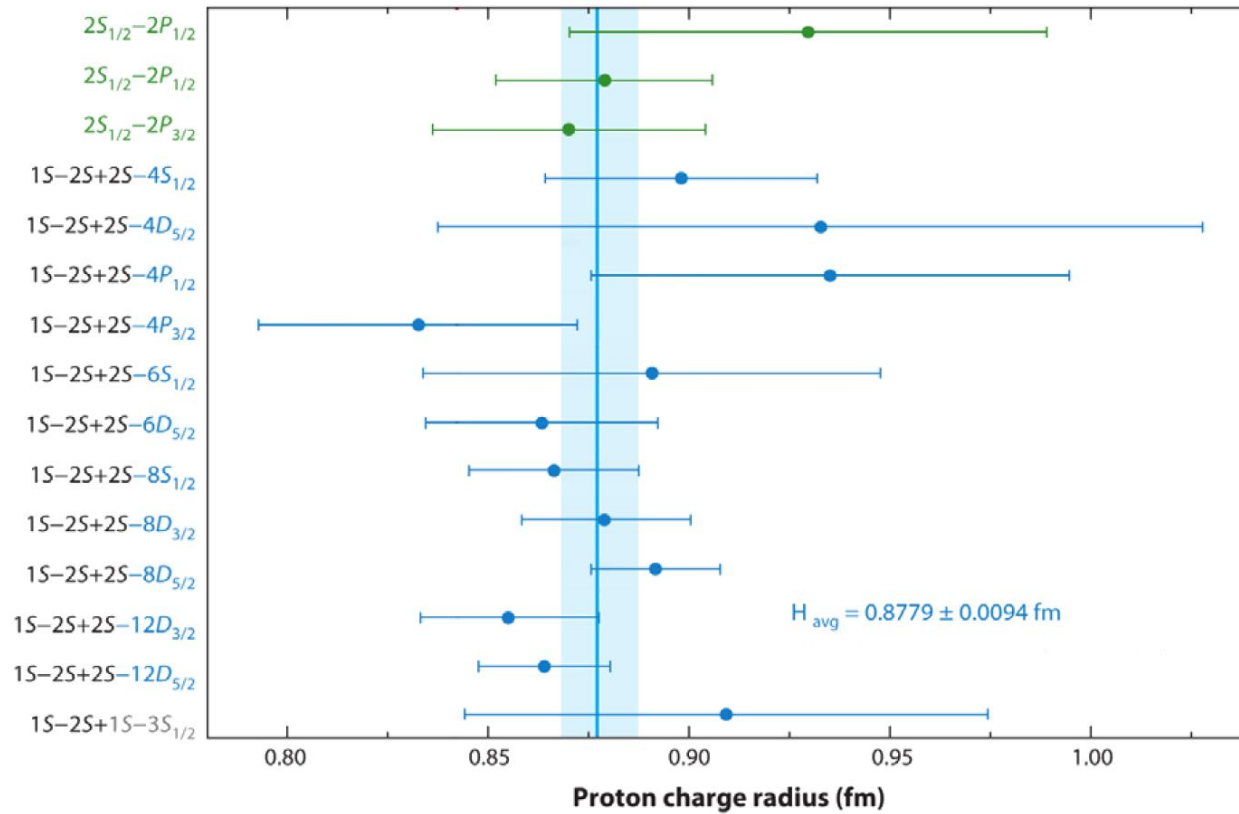
- Discovery of proton's finite size and charge radius
- Two methods for measuring the proton charge radius
- Experiments for both methods
- Results and the “proton radius puzzle”
- Possible solutions

A series of thin, light brown lines forming an abstract geometric pattern on the left side of the slide. The lines intersect to create various polygons and shapes, extending from the top left towards the bottom left.

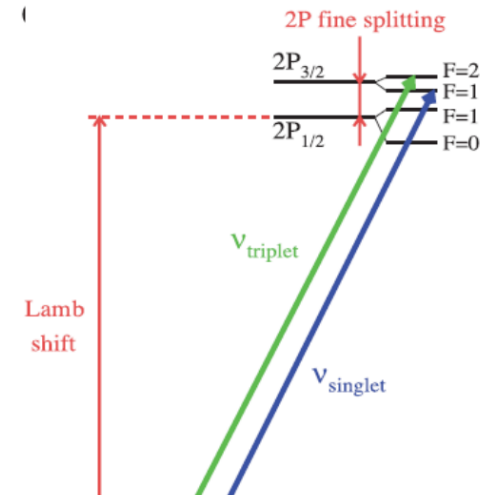
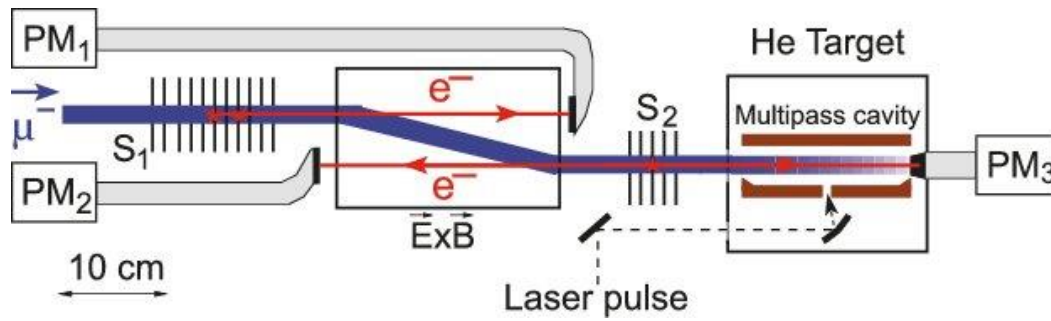
THANK YOU

Questions?

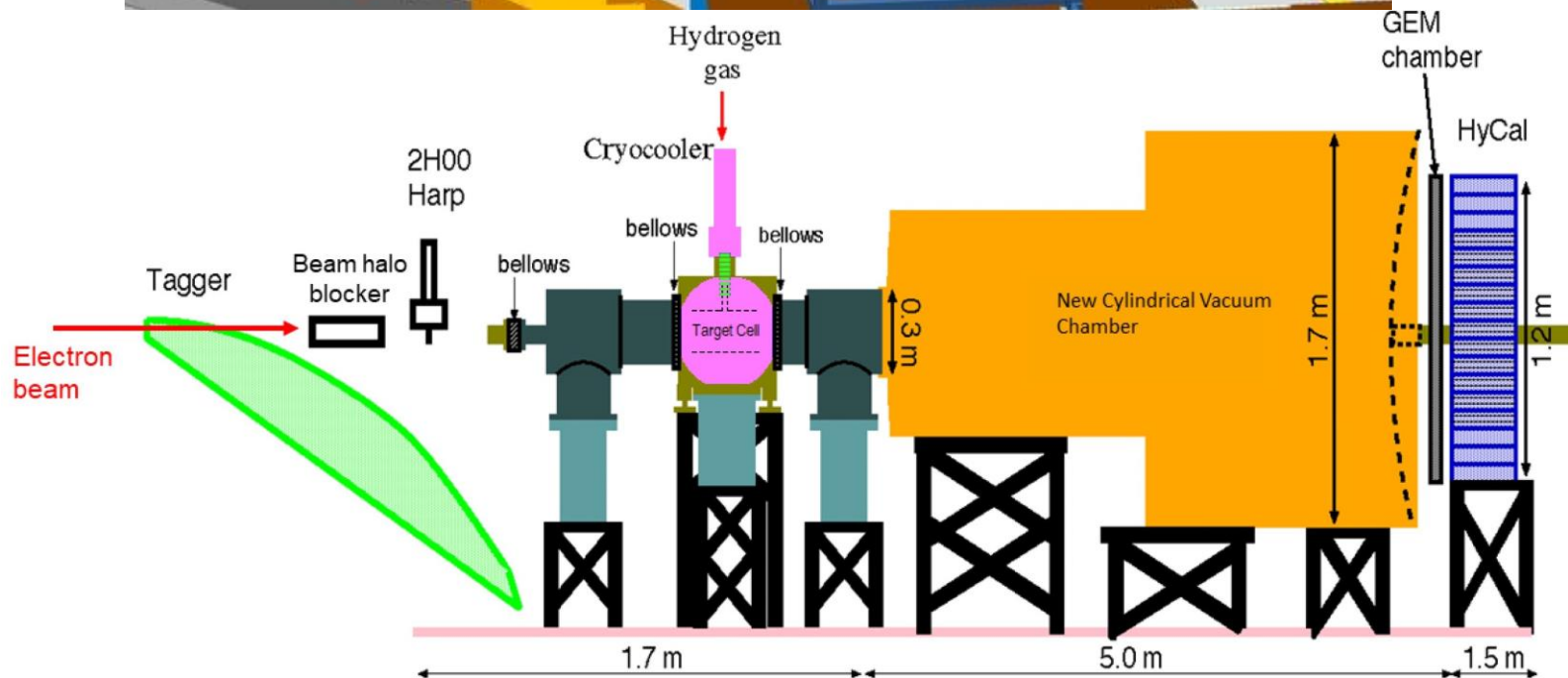
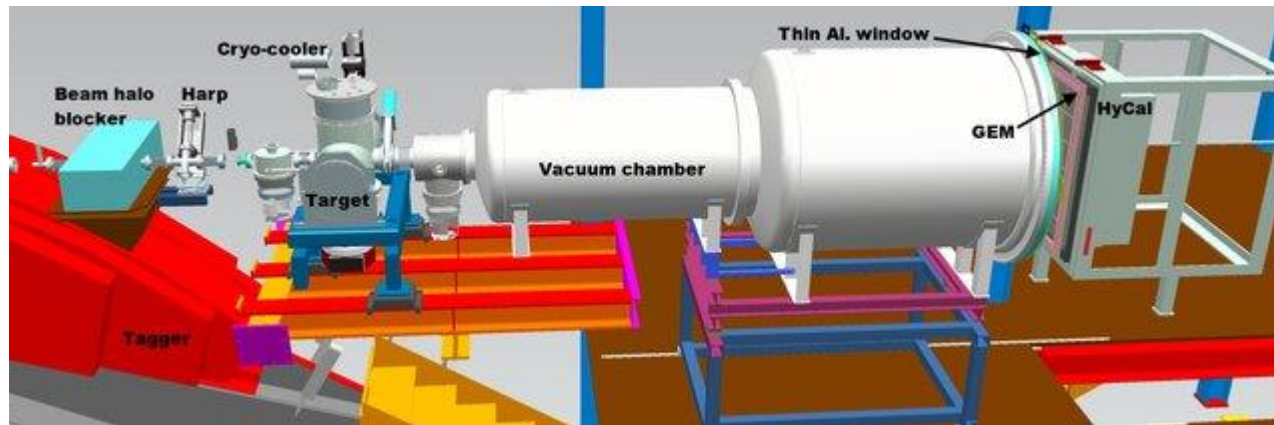
BACKUP SLIDE



BACKUP SLIDE



BACKUP SLIDE



THE THEORY OF ELASTIC SCATTERING

- Alpha $\rightarrow$ electron (EM interaction)
- Scattering experiments measure differential cross-section:

$$d\sigma = \frac{|\mathcal{M}|^2}{F} dQ, \quad -i\mathcal{M} = j^\mu \left(-\frac{g_{\mu\nu}}{q^2} \right) J^\nu,$$

- Electron current:

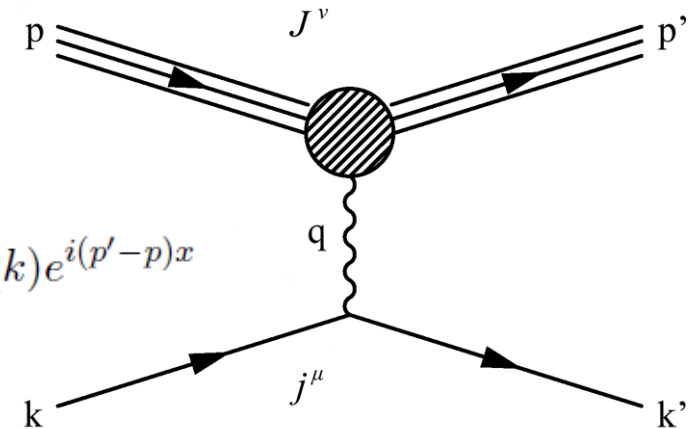
$$j^\mu = -e\bar{u}(k')\gamma^\mu u(k)e^{i(k-k')x}$$

- Proton current:

$$J^\mu = -e\bar{u}(p') \left[F_1(q^2)\gamma^\mu + \frac{\kappa}{2M} F_2(q^2) i\sigma^{\mu\nu} q_\nu \right] u(k) e^{i(p'-p)x}$$

- Differential cross-section:

$$\left(\frac{d\sigma}{d\Omega} \right)_{lab} = \left(\frac{\alpha^2}{4E^2 \sin^4 \frac{\theta}{2}} \right) \frac{E'}{E} \left[(F_1^2 - \frac{\kappa^2 q^2}{4M^2} F_2^2) \cos^2 \frac{\theta}{2} - \frac{q^2}{2M^2} (F_1 + \kappa F_2)^2 \sin^2 \frac{\theta}{2} \right]$$



THE THEORY OF ELASTIC SCATTERING

- Electric and magnetic form factor:

$$G_E(Q^2) = F_1(Q^2) + \frac{\kappa q^2}{4M^2} F_2(Q^2),$$

$$G_M(Q^2) = F_1(Q^2) + \kappa F_2(Q^2).$$

- The Rosenbluth formula:

$$\left(\frac{d\sigma}{d\Omega} \right)_{lab} = \left(\frac{\alpha^2}{4E^2 \sin^4 \frac{\theta}{2}} \right) \frac{E'}{E} \left[\frac{G_E^2 + \tau G_M^2}{1 + \tau} \cos^2 \frac{\theta}{2} - 2\tau G_M^2 \sin^2 \frac{\theta}{2} \right]$$

- Electric and magnetic charge radius:

$$R_E^2 = -6 \frac{dG_E}{dQ^2} \bigg|_{Q^2=0}, \quad R_M^2 = -6 \frac{dG_M}{dQ^2} \bigg|_{Q^2=0}.$$

BACKUP SLIDE

